

For large n , $\sqrt{2\chi^2}$ can be shown to be approximately normally distributed with mean $\sqrt{2n-1}$ and standard deviation 1. This approximation is generally used to calculate values of χ^2 at different probability levels for $n > 30$.

Let us denote $\chi_{\alpha,n}^2$, the value of χ^2 (with $df = n$) for which $p[\chi^2 > \chi_{\alpha,n}^2] = \alpha$ i.e., the area to the right of the point $\chi_{\alpha,n}^2$ is α (see Fig. 29.2).

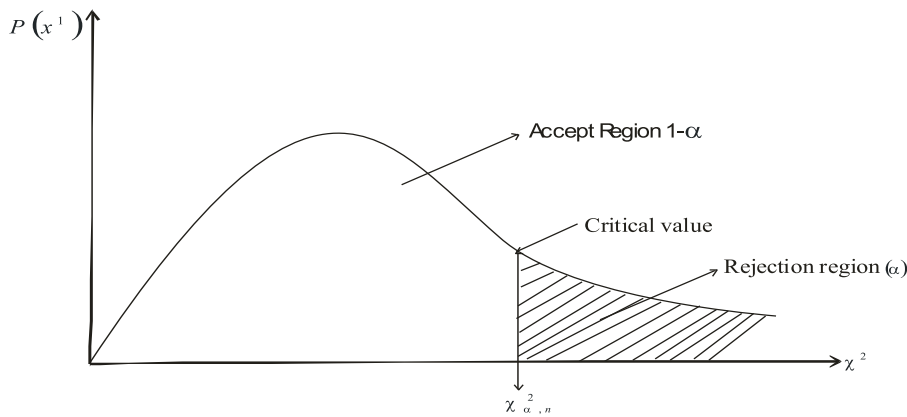


Fig. 29.2: Critical Value of χ^2 Distribution

The value $\chi_{\alpha,n}^2$ is known as upper (right - tailed) α point or Critical Value or Significant Value of χ^2 for n df and has been tabulated for different values of n and α in the χ^2 Table. From the Table, we observe that the critical values of χ^2 increase as $n(df)$ increases and level of significance (α) decreases.

Note: The lower (left-tailed) α point of the χ^2 distribution is given by $\chi_{1-\alpha,n}^2$.

If X has a standard normal distribution, then X^2 has the Chi-square distribution with $(n-1)$ degree of freedom. In other words, if $X_1, X_2, X_3, \dots, X_n$ are independent random variables having standard normal distribution, then $Y = \sum_{i=1}^n X_i^2$ has the

Chi-square distribution with $n-1$ degrees of freedom.

Proof: Using moment generating function (discussed in the previous unit) with $df = 1$, we find that:

$$M_{X_i^2}(t) = (1 - 2t)^{-\frac{1}{2}}$$

$$\text{It follows that: } M_{X_i^2}(t) = \prod_{i=1}^n (1 - 2t)^{-\frac{1}{2}} = (1 - 2t)^{-\frac{n}{2}}$$

Thus, moment generating function is readily identified as the Chi square distribution with n degrees of freedom.

Following are two important properties of Chi Square distribution:

- 1) Let $X_1, X_2, X_3, \dots, X_n$ be n random variables having Chi square distributions with $n_1, n_2, n_3, \dots, n_n$ degrees of freedom. Then, if X_1, X_2 are two independent random variables, with X_1 having a Chi-square distribution with n_1 degrees of freedom, and $X_1 + X_2$ having a Chi-square distribution with n ($> n_1$) degrees of freedom, then X_2 has a Chi-square distribution with $n - n_1$ degrees of freedom.
- 2) If $\bar{X}$ and S^2 are the mean and the variance of a random sample of size n from a normal population with a mean μ and the standard deviation σ , then
 - a) $\bar{X}$ and S^2 are independent and
 - b) The random variable $\frac{(n-1)S^2}{\sigma^2}$ has a Chi-square distribution with $(n-1)$ degrees of freedom.

Proof of the 2nd part:

We begin with the identity:

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \bar{X} + \bar{X} - \mu)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$$

If we divide each term by σ^2 and substitute $(n-1)S^2$ for $\sum_{i=1}^n (X_i - \bar{X})^2$, we get:

$$\sum_{i=1}^n (X_i - \mu)^2 = \frac{(n-1)S^2}{\sigma^2} + \left\{ \frac{(\bar{X} - \mu)^2}{\frac{\sigma}{\sqrt{N}}} \right\}^2$$

Left hand side of the above identity is representing a random variable with Chi-square distribution with n degrees of freedom, whereas the second term of the right-hand side is a random variable with Chi-square distribution with 1 degree of freedom. Since $\bar{X}$ and S^2 are assumed to be independent, it follows that the two terms on the RHS of the equation are independent and we conclude that $\frac{(n-1)S^2}{\sigma^2}$ is a random variable having a Chi-square distribution with $(n-1)$ degrees of freedom.

29.3.1.1 Chi-square Test of Goodness of Fit

'Chi-square test of goodness of fit' is a test for testing the significance of the discrepancy between theory and experiment.

If O_i , ($i = 1, 2, \dots, n$) is a set of observed (experimental frequencies) and E_i ($i = 1, 2, \dots, n$) is the corresponding set of expected (theoretical or hypothetical)

frequencies, then Karl Pearson's chi-square test for goodness of fit is given by:

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right], \text{ where } \left(\sum_{i=1}^n O_i = \sum_{i=1}^n E_i \right)$$

χ^2 above follows a chi-square distribution with (n-1) df. This is an approximate test for large values of n.

29.3.2 't' Distribution

29.3.2.1 Student's 't'

Let x_i , ($i = 1, 2, \dots, n$) be a random sample of size n from a normal population with mean μ and variance σ^2 . Then the random variable $\bar{X}$ has a normal distribution with mean μ and the variance $\frac{\sigma^2}{n}$. In other words, $\frac{(\bar{X} - \mu)}{\frac{\sigma}{\sqrt{n}}}$ has the

standard normal distribution. Although this is an important result, there is a major difficulty in applying it when the population standard deviation σ is unknown. This makes it necessary to replace σ by sample standard deviation S . Then student's 't' is defined by the statistic

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is the sample mean and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimate of the population variance σ^2 ,

and it follows t distribution with $v = (n - 1)$ df, with probability density function

$$f(t) = \frac{1}{\sqrt{v} B\left(\frac{1}{2}, \frac{v}{2}\right)} \cdot \frac{1}{\left[1 + \frac{t^2}{v}\right]^{(v+1)/2}}; -\infty < t < \infty$$

A statistic t following Student's t - distribution with n df will be abbreviated as $t \sim t_n$.

If we take $v = 1$, then $f(t) = \frac{1}{B\left(\frac{1}{2}, \frac{1}{2}\right)} \cdot \frac{1}{(1 + t^2)}$

$$= \frac{1}{\pi} \cdot \frac{1}{(1 + t^2)}; -\infty < t < \infty$$

$$\left[\therefore \Gamma\left(\frac{1}{2}\right) \sqrt{\pi} \right]$$

which is the p.d.f of standard Cauchy distribution.

Hence, when $v = 1$, Student's t distribution reduces to Cauchy distribution.

29.3.2.2 Fisher's 't'

It is the ratio of a standard normal variate to the square root of an independent chi-square variate divided by its degree of freedom. If τ is a $N(0, 1)$ and χ^2 is an independent chi-square variate with n degrees of freedom, then Fisher's t is given by

$$t = T / \sqrt{\chi^2 / n}$$

and it follows Student's t distribution with n degrees of freedom.

Since τ and χ^2 are independent, their joint density function is given by

$$\frac{1}{\sqrt{2\pi}} \cdot \exp\left(-T^2/2\right) \cdot \frac{\exp(-\chi^2/2)(\chi^2)^{\frac{n}{2}-1}}{2^{n/2} \cdot \Gamma(n/2)},$$

$$-\infty < T < \infty, 0 < \chi^2 < \infty.$$

Making one-to-one transformation

$$\left. \begin{array}{l} t = T / \sqrt{\chi^2 / n} \\ u = \chi^2 \end{array} \right\} (-\infty < t < \infty, 0 < u < \infty),$$

so that $T = t\sqrt{u/n}$, $\chi^2 = u$,

and noting that the Jacobian of the transformation is

$$J = \begin{vmatrix} \frac{\partial T}{\partial t} & \frac{\partial T}{\partial u} \\ \frac{\partial \chi^2}{\partial t} & \frac{\partial \chi^2}{\partial u} \end{vmatrix} = \begin{vmatrix} \sqrt{u/n} & \frac{1}{2\sqrt{un}} \\ 0 & 1 \end{vmatrix} = \sqrt{u/n}$$

The joint p.d.f. of t and u becomes

$$\frac{1}{\sqrt{2\pi} \cdot 2^{\frac{n}{2}} \cdot \Gamma(n/2) \cdot \sqrt{n}} \exp\left\{-\frac{u}{2} \left(1 + \frac{t^2}{n}\right)\right\} u^{\frac{n}{2}-1}$$

$$-\infty < t < \infty, 0 < u < \infty$$

The p.d.f. of t is, therefore,

$$\frac{1}{\sqrt{2\pi} \cdot 2^{n/2} \cdot \Gamma(n/2) \cdot \sqrt{n}} \left[\int_0^\infty \exp\left\{-\frac{u}{2} \left(1 + \frac{t^2}{n}\right)\right\} u^{(n-1)/2} du \right] dt$$

$$= \frac{1}{\sqrt{2\pi} \cdot 2^{n/2} \cdot \Gamma(n/2)} \cdot \frac{\Gamma[(n+1)/2]}{\left[\frac{1}{2} \left(1 + \frac{t^2}{n}\right)\right]^{(n+1)/2}}$$

$$= \frac{\Gamma[(n+1)/2]}{\sqrt{n} \Gamma(n/2) \Gamma\left(\frac{1}{2}\right)} \cdot \frac{1}{\left[1 + \frac{t^2}{n}\right]^{(n+1)/2}}$$

$$-\alpha < t < \alpha$$

$$= \frac{1}{\sqrt{nB}\left(\frac{1}{2}, \frac{n}{2}\right) \left[1 + \frac{t^2}{n}\right]^{(n+1)/2}}, -\alpha < t < \alpha$$

which is same as the probability function of Student's t - distribution with n df. It should be noted here that, Student's t is a particular case of Fisher's ' t '. Like the standard normal distribution, the t distribution is symmetrical about $t = 0$. But unlike the normal distribution, it is more peaked than a normal distribution with the same standard deviation (Fig. 29.3).

The symbol $t_{\alpha,n}$ will be used to denote the value of t (with $df = n$) such that

$$p[t > t_{\alpha,n}] = \alpha$$

$$t_{1-\alpha,n} = -t_{\alpha,n}.$$

For small n , the t distribution differs considerably from the standard normal distribution, $t_{\alpha,n}$ being always greater than T_{α} if $0 < \alpha < \frac{1}{2}$. For large value of n , however, the t distribution tends to the standard normal form and $t_{\alpha,n}$ may then be well approximated by T_{α} .

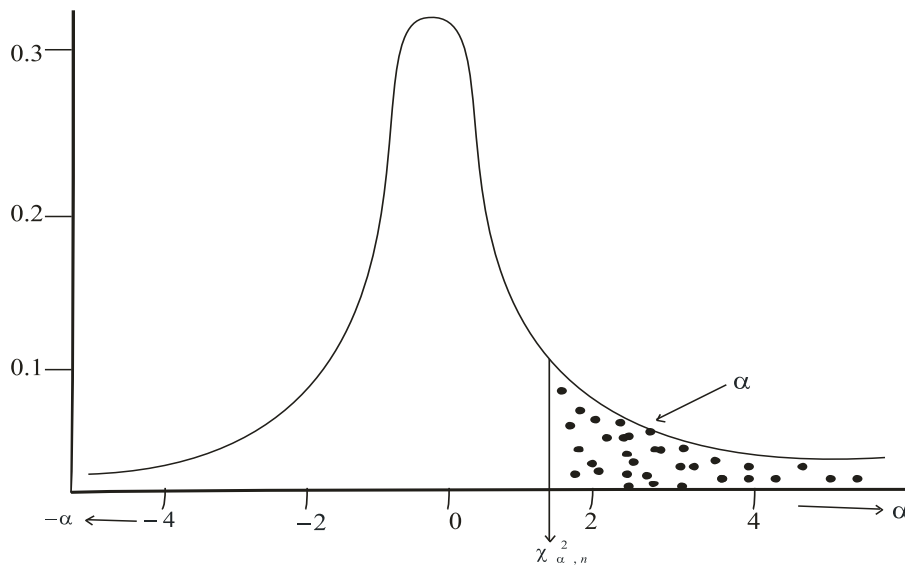


Fig. 29.3: t distribution with 5 Degrees of Freedom ($n=5$)

29.3.3 'F' Distribution

If X and Y are two independent chi-square variates with n_1 and n_2 df respectively, then F statistic is defined by

$$F = \frac{X/n_1}{Y/n_2}$$

In other words, F is defined as the ratio of two independent chi-square variates divided by their respective degrees of freedom and it follows Snedecor's F distribution with (n_1, n_2) df with probability function given by

$$\frac{\left(\frac{n_1}{n_2}\right)^{n_1/2}}{B(n_1/2, n_2/2)} \frac{F^{\frac{n_1}{2}-1}}{\left[1 + \frac{n_1}{n_2} F\right]^{\left(\frac{n_1+n_2}{2}\right)}}, 0 \leq F < \infty$$

Note: i) The sampling distribution of F-statistic does not involve any population parameters and depends only on the degrees of freedom n_1 and n_2 .

ii) A statistic F following Snedecor's F distribution with (n_1, n_2) df will be denoted by $F \sim F(n_1, n_2)$.

To derive the above result, note that the joint p.d.f of X and Y, is given by:

$$g(x, y) = \frac{1}{2^{(n_1+n_2)/2} \Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right)} \cdot X^{(n_1/2)-1} \cdot Y^{\left(\frac{n_2}{2}\right)-1} \cdot \exp[-(X+Y)/2]$$

$$0 < X < \alpha, 0 < Y < \alpha.$$

Let us now make the one-to-one transformation

$$\begin{cases} F = \frac{X/n_1}{Y/n_2} \\ u = Y \end{cases} \left\{ \begin{array}{l} 0 < F < \alpha, 0 < u < \alpha \end{array} \right.$$

$$\text{So that, } X = \frac{n_1}{n_2} Fu.$$

$$\text{and } Y = u.$$

The Jacobian of the transformation is

$$J = \begin{vmatrix} \frac{\partial X}{\partial F} & \frac{\partial X}{\partial u} \\ \frac{\partial Y}{\partial F} & \frac{\partial Y}{\partial u} \end{vmatrix} = \begin{vmatrix} \frac{n_1}{n_2} u & \frac{n_1}{n_2} F \\ 0 & 1 \end{vmatrix} = \frac{n_1}{n_2} u$$

Hence, the joint p.d.f of F and u is

$$\begin{aligned} h(F, u) &= \frac{\left(\frac{n_1}{n_2}\right)^{\left(\frac{n_1}{2}\right)}}{2^{(n_1+n_2)/2} \Gamma(n_1/2) \Gamma(n_2/2)} \cdot u^{[(n_1+n_2)/2]-1} \cdot F^{(n_1/2)-1} \\ &\times \exp\left[\frac{-u}{2} \left(1 + \frac{n_1}{n_2} F\right)\right] \end{aligned}$$

$$0 < F < \alpha, 0 < u < \alpha.$$

The p.d.f of F is therefore,

$$f(F) = \int_0^d h(F, u) du$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{(n_1/2)}}{\Gamma\left(\frac{n_1}{n_2}\right)\Gamma\left(\frac{n_2}{2}\right)} \times F^{\left(\frac{n_1}{2}\right)-1} \times \frac{\Gamma\left(\frac{n_1+n_2}{2}\right)}{\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{(n_1/2)}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \times \frac{F^{\left(\frac{n_1}{2}\right)-1}}{\left(1+\frac{n_1}{n_2}F\right)^{(n_1+n_2)/2}}, 0 < F < \infty$$

The distribution is highly positively skewed (Fig. 29.4). It is easily seen from the definition of t and F that an F with $n_1 = 1$ is a t^2 , t having $df = n_2$.

As in the previous cases, we shall denote by $F_{\alpha;n_1,n_2}$ the upper α point of the F distribution with $df = (n_1, n_2)$; i.e.,

$$p[F > F_{\alpha;n_1,n_2}] = \alpha$$

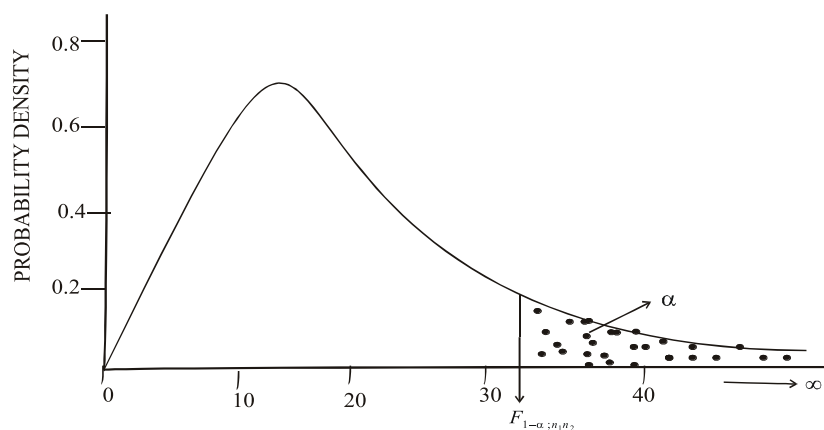


Fig. 29.4: F distribution with (10, 4) Degrees of Freedom ($n_1=10, n_2=4$)

As regards the lower α point $F_{1-\alpha;n_1,n_2}$, we see that

$$p[F < F_{1-\alpha;n_1,n_2}] = \alpha$$

$$\text{or, } p\left[\frac{1}{F} > \frac{1}{F_{1-\alpha;n_1,n_2}}\right] = \alpha$$

Now, $\frac{1}{F}$, which is of the form $\frac{Y/n_2}{X/n_1}$, is itself distributed as an F with $df = (n_2, n_1)$.

It follows that

$$\frac{1}{F_{1-\alpha;n_1,n_2}} = F_{1-\alpha;n_2,n_1}$$

$$F_{1-\alpha;n_1,n_2} = \frac{1}{F_{\alpha;n_2,n_1}}$$

It is, therefore, unnecessary to tabulate the lower α points of F distributions with various degrees of freedom, once the upper α points are tabulated.

Check Your Progress 1

- 1) Discuss the features of χ^2 , t and f distributions.
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- 2) A die is suspected of being biased. It is rolled 24 times with the following result.

Outcome	1	2	3	4	5	6
Frequency	8	4	1	8	3	0

Conduct a significance test to see if the die is biased.

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- 3) When can you use a χ^2 or a z test and reach the same conclusion?
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- 4) The following figures show the distribution of digits in numbers chosen at random from a telephone directory.

Digits:	0	1	2	3	4	5	6	7	8	9	Total
Frequency:	1026	1107	997	966	1075	933	1107	972	964	853	10,000

Test whether the digits may be taken to occur equally frequently in the directory.

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29.4 SAMPLING DISTRIBUTIONS OF MEAN AND VARIANCE IN RANDOM SAMPLING FROM A NORMAL DISTRIBUTION

Since statistics are random variables, their values will vary from sample to sample and the distribution is referred to as sampling distribution.

Let $x_1, x_2, \dots, x_n$ be a random sample from a normal distribution whose p.d.f is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], -\infty < x < \infty$$

We shall denote the sample mean and the sample variance of x by $\bar{x}$ and s'^2 , respectively.

Thus,
$$\bar{x} = \sum_{i=1}^n x_i / n$$

and
$$s'^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

In order to obtain the sampling distributions of $\bar{x}$ and s'^2 , we start from the joint p.d.f of $x_1, x_2, \dots, x_n$, which is

$$\pi_i f(x_i) = \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left[-\sum_i (x_i - \mu)^2 / 2\sigma^2\right]$$

We make the following one-to-one transformation from x_i ($i = 1, 2, \dots, n$) to y_i ($i = 1, 2, \dots, n$):

$$y_1 = \frac{(x_1 - \mu)/\sigma + (x_2 - \mu)/\sigma + \dots + (x_n - \mu)/\sigma}{\sqrt{n}},$$

$$y_i = a_{i1}\left(\frac{(x_1 - \mu)}{\sigma}\right) + a_{i2}\left(\frac{(x_2 - \mu)}{\sigma}\right) + \dots + a_{in}\left(\frac{(x_n - \mu)}{\sigma}\right), \text{ for } i = 2, 3, \dots, n.$$

where the $(n-1)$ vectors $(a_{i1}, a_{i2}, \dots, a_{in})$ are of unit length, mutually orthogonal and each orthogonal to the vector $\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \dots, \frac{1}{\sqrt{n}}\right)$.

One such set of vectors is:

$$\frac{1}{\sqrt{2}} (1, -1, 0, 0, \dots, 0, 0),$$

$$\frac{1}{\sqrt{6}} (1, 1, -2, 0, \dots, 0, 0)$$

$$\frac{1}{\sqrt{n(n-1)}} (1, 1, 1, \dots, 1, -(n-1)) \quad .$$

The Jacobian of the transformation is then J , such that

$$\frac{1}{J} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \dots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \dots & \frac{\partial y_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \frac{\partial y_n}{\partial x_2} & \dots & \frac{\partial y_n}{\partial x_n} \end{vmatrix} = \mp \frac{1}{\sigma^n},$$

implying that $J = \mp \sigma^n$ and $|J| = \sigma^n$.

$$\text{Further, } \sum_{i=1}^n y_i^2 = \sum_{i=1}^n (x_i - \mu)^2 / \sigma^2$$

Hence, the joint p.d.f of $y_1, y_2, \dots, y_n$ is

$$\frac{1}{(\sqrt{2\pi})^n} \exp \left[-\sum_{i=1}^n y_i^2 / 2 \right]$$

This shows that $y_1, y_2, \dots, y_n$ are independently and identically distributed, each being a standard normal variable.

But $y_i = \frac{1}{\sigma\sqrt{n}} \sum_i (x_i - \mu) = \frac{\sqrt{n}(\bar{x} - \mu)}{\sigma}$, is a linear function of $\bar{x}$. Since y_1 is a standard normal variable, $\bar{x}$ must be a normal variable with mean μ and variance $\frac{\sigma^2}{n}$.

Thus, the p.d.f of $\bar{x}$ is

$$g(\bar{x}) = \frac{\sqrt{n}}{\sigma\sqrt{2\pi}} \exp \left[-n(\bar{x} - \mu)^2 / 2\sigma^2 \right], -\infty < \bar{x} < \infty$$

$$\begin{aligned} \text{Again, } \sum_{i=2}^n y_i^2 &= \sum_{i=1}^n y_i^2 - \bar{y}^2 \\ &= \frac{1}{\sigma^2} \left[\sum_{i=1}^n (x_i - \mu)^2 - n(\bar{x} - \mu)^2 \right] \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{(n-1)s'^2}{\sigma^2} \end{aligned}$$

Now, $\sum_{i=1}^n y_i^2$, being the sum of squares of $(n-1)$ independent standard normal variables, is a χ^2 with $df = (n-1)$, and this is distributed independently of y_1 . It follows that $(n-1)s'^2 / \sigma^2$ is distributed as a χ^2 with $df = (n-1)$ and is independent of $\bar{x}$.